

Overprocessing in neuroimaging processing pipelines, illustrated with functional Near-Infrared Spectroscopy (fNIRS)

Felipe Orihuela-Espina^a, Robert Ward^a

^aUniversity of Birmingham, School of Computer Science, Edgbaston, Birmingham, United Kingdom

Abstract. Overprocessing occurs when we exceed reasonable processing to extract information from observations. Overprocessing can severely affect interpretation of results, e.g. increasing false positives. This paper introduces the concept of overprocessing and its associated risk to the neuroimaging community. The theoretical underpinnings revealing the existence of the problem are given, and the problem is formally stated. The case is exemplified using fNIRS. The existence of an operation equivalent to some arbitrary processing and analysis pipeline capable of systematically projecting any experimental observation $\mathbf{x}_i$ in neuroimaging into a discretionary hypothesis $\mathbf{x}_h$ is theoretically demonstrated and empirically evidenced over both synthetic examples and experimental data. The transfer function is discussed as a plausible non-trivial compliant operation. The analysis of both this transfer function and the problem geometry are discussed as potential ways to constraint the problem. At present, the neuroimaging community lacks criteria to alleviate the risk of overprocessing. This paper intends to raise awareness on this largely unknown issue.

Keywords: fNIRS, overprocessing, data analysis, signal processing, neuroimaging.

*Felipe Orihuela-Espina, f.orihuela-espina@bham.ac.uk

Coming soon!